TOPS Technologies

Interview Questions — DA - Working with Pandas

Total Questions: 44

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1. What is Pandas, and why is it useful in data analysis?

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2. Explain the difference between a Pandas Series and a DataFrame.

> **Hint:** Think about the number of dimensions each structure represents.

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3. How would you manually create a simple DataFrame with three columns: 'Name', 'Age', and 'City' in Pandas?

> **Hint:** Consider using the pd.DataFrame constructor with a dictionary.

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4. How can you read a CSV file into a Pandas DataFrame?

> **Hint:** There is a specific Pandas function for reading CSV files.

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5. What is the difference between the head() and tail() methods in Pandas?

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6. If you want to get a quick statistical summary of all numeric columns in your DataFrame, which method would you use?

> **Hint:** This method gives you count, mean, std, min, and max values.

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7. How would you export a DataFrame to an Excel file?

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8. How do you select a specific column from a DataFrame?

> **Hint:** You can use either dot notation or bracket notation.

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9. What is the difference between loc and iloc in Pandas?

> **Hint:** Think about whether you are selecting by label or by position.

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10. Write a Pandas statement to select all rows where the 'Revenue' column is greater than 10,000.

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11. How would you sort a DataFrame by the 'Price' column in descending order?

> **Hint:** Look for a method that sorts values and allows you to specify ascending or descending.

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12. What does resetting the index of a DataFrame do, and when might you need to use it?

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13. If you want to reindex a DataFrame to match a new set of row labels, which method would you use?

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14. How can you check for missing values in a DataFrame?

> **Hint:** There is a method that returns True for missing entries.

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15. What is the difference between dropna() and fillna() in Pandas?

> **Hint:** One removes missing data, the other replaces it.

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16. How would you fill missing values in the 'Age' column with the median age?

> **Hint:** You need to combine fillna() with another method to get the median.

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17. What function would you use to remove duplicate rows from a DataFrame?

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18. Explain the concept of outliers and how the IQR method helps in detecting them.

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19. How can you use the Z-score to identify outliers in a numeric column?

> **Hint:** Z-score measures how many standard deviations a value is from the mean.

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20. How would you add a new column 'Revenue' to a DataFrame, calculated as 'Price' multiplied by 'Qty'?

> **Hint:** You can create new columns by assigning a calculation to a new column name.

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21. What is the difference between apply() and applymap() in Pandas?

> **Hint:** Consider whether the operation is applied to a Series or to every element in the DataFrame.

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22. Write a Pandas expression using apply and a lambda function to increase all values in the 'Discount' column by 10%.

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23. How would you convert all values in a 'Category' column to lowercase in Pandas?

> **Hint:** Look for string methods accessible via .str.

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24. Which Pandas method would you use to check if a string column contains a specific substring?

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25. How can you split a column containing 'FirstName LastName' into two separate columns?

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26. How do you convert a column of string dates into Pandas datetime objects?

> **Hint:** There is a function that starts with pd.to_

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27. How would you extract the year and month from a Pandas datetime column?

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28. Suppose you have a 'SubscriptionDate' column; how would you calculate the tenure (in days) for each customer?

> **Hint:** Subtract the subscription date from today's date, and use the dt accessor.

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29. What is the difference between merge() and join() in Pandas?

> **Hint:** Think about whether you are joining on indexes or columns.

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30. How would you perform an inner join between two DataFrames on the columns 'CustomerID' and 'OrderID'?

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31. Explain the difference between left, right, inner, and outer joins in Pandas.

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32. How can you concatenate two DataFrames row-wise and column-wise?

> **Hint:** There is a specific method that can take an axis parameter.

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33. What is the purpose of the pivot() function in Pandas, and how does it differ from pivot_table()?

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34. How would you combine monthly sales files (each as a DataFrame) into a single DataFrame for analysis?

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35. Explain how groupby() works in Pandas and give an example use case.

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36. How would you calculate the total revenue by region and category using groupby and agg()?

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37. What is a multi-aggregation in Pandas groupby, and how would you apply it?

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38. Describe the steps you would take to perform univariate and bivariate analysis using Pandas.

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39. How can you check the correlation between two numeric columns in a DataFrame?

> **Hint:** There is a method that returns the Pearson correlation coefficient.

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40. Suppose you are given a raw dataset. Walk me through the steps you would take in Pandas to clean, transform, merge, and analyze the data to answer business questions.

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41. How would you use Pandas to generate insights from a case study dataset, such as identifying top-performing products or regions?

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42. If you were using ChatGPT or a similar AI tool to help you write a complex Pandas data transformation, how would you ensure the suggested code is correct and safe to use?

> **Hint:** Think about testing, understanding the code, and checking for errors.

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43. How could AI-powered tools assist you in performing Exploratory Data Analysis (EDA) with Pandas, and where would you still rely on your own data understanding?

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44. You need to merge customer and sales data, clean missing values, and create a pivot table to summarize sales by category. Briefly describe how you would integrate these tasks in a Pandas workflow.
